

## MATLAB SESSION-05

1. Find the laplace transformation of  $e^{(at)} * t^n$ .

**Output:**  $n! / (s-a)^{n+1}$ .

2. Plot a 3D graph of a given Laplace Transform of a function.  $f(t) = (\sin^3 * (2t))$ .

$$F(s) = 48 / ((s^2 + 4)(s^2 + 36)).$$

3. Find  $L(e^{(3t)} \sqrt{3*t})$ .

**Output:**  $(\sqrt{\pi}/2) * (1/(s-3)^{3/2})$